Hardware constraints for a QEC Transformer

Technical expansion of the eight-page Hardware Constraints Notebook (in Progress) | FPGA inference at 4 K | Latency, power, architecture, and training-memory distinctions

Source and scope. Sections 1–3 expand the notebook's timing and Transformer notes (pages 1–3); Section 4 covers complete-window and streaming decoding (page 4); Sections 5–6 expand cryogenic power and architecture choices (pages 5–6); Section 7 covers training batches and architecture-dependent memory (pages 7–8). Equations and external research provide technical expansion. No FPGA part or cooling allocation is specified, so no measured project-specific wattage or latency is claimed.

1. Inference latency: physical meaning and deadlines

Inference latency is the time from the arrival of the final required syndrome input at the deployed FPGA to a valid predicted logical correction. It limits how rapidly QEC information can enter the classical control loop. Delayed decisions may create backlog or miss a deadline for conditional logical operations. Not every measured error needs immediate physical correction: many systems record Pauli-frame updates, but real-time decoding still needs adequate throughput and deadline compliance.

Timing measure	Definition	Why it matters
Inference latency, L	Input-available-to-output time for one decoded prediction.	Response-time deadline.
Initiation interval, II	Minimum interval between the start of successive predictions.	Sustainable throughput.
End-to-end feedback	Measurement classification, communication, buffering, inference and output delivery.	Time correction information actually reaches the controller.

$$T_{feedback} = T_{acquisition} + T_{transfer} + T_{buffer} + L + T_{delivery}$$

For a new syndrome update every T_{round}, an implementation that starts one inference per update generally needs $II \le T_{round}$ to avoid an accumulating queue. A pipelined decoder could have L = 2 µs and II = 0.5 µs, so it keeps pace with 1 µs updates but cannot satisfy a strict 1 µs response deadline for any individual update.

1.1 Clock cycle budget

$$T_{clk} = 1/f_{clk} \; ; \; L = N_{cycles}/f_{clk} \; ; \; N_{max} = \mathrm{floor}(L_{target}\ f_{clk})$$

FPGA clock	Period	Cycles in 1 µs
50 MHz	20 ns	50
100 MHz	10 ns	100
200 MHz	5 ns	200
250 MHz	4 ns	250

FPGA clock	Period	Cycles in 1 µs
400 MHz	2.5 ns	400

At 100 MHz, a 100-cycle datapath takes 1 µs. A 135-cycle datapath takes 1.35 µs, exceeding a strict compute-only target. The clock rate is constrained by post-route logic delay and interconnect, not merely by the rate requested in a hardware design file.

2. Transformer inference: operations and dependency chain

Define T = token count, m = detector features per token, d = embedding dimension, H = attention heads, d_h = d/H and f = feedforward width. Each step maps onto FPGA arithmetic units, registers, memory and routing. In a round-token representation, a token contains measurements associated with a defined time slice; detector coordinates and availability should be preserved.

2.1 Embedding and projections

$$E = XW_{emb} + b_{emb} + P \; ; \; X \in \{0,1\}^{T \times m} \; ; \; MAC_{embed} \approx T\,m\,d$$

$$Q = E\,W_Q \; ; \; K = E\,W_K \; ; \; V = E\,W_V$$

Binary 0/1 input does not constrain the learned latent representation to arithmetic in the two-element field. A projection maps detector bits into real-valued or fixed-point features. In custom hardware, multiplication by a binary input can be simplified to a conditional add; non-firing detectors remain informative through the difference between input patterns, learned biases and positional features.

Q, K and V are independent once E exists, so their projections may be run concurrently if adequate DSP capacity and memory bandwidth exist. With a shared datapath, the same arithmetic units are reused, increasing the schedule length.

2.2 Attention scores and value aggregation

$$S_h = Q_h\,K_h^T/\mathrm{sqrt}(d_h) \; ; \; A_h = \mathrm{softmax}(S_h) \; ; \; O_h = A_h\,V_h$$

$$MAC_{QKV\,+\,output} \approx 4Td^2 \; ; \; MAC_{attention} \approx 2T^2d$$

Attention scoring must wait until Q and K are available; aggregation must wait for the attention weights and V. Even massive parallelism cannot remove these logical dependencies. With fixed total d, increasing the head count does not automatically increase the leading-order MAC count, although it can complicate buffers and scheduling.

2.3 Softmax, normalization and feedforward

$$\mathrm{softmax}(s_j) = \exp(s_j) \,/\, \Sigma_k \exp(s_k)$$

$$FFN(x) = W_2\,\sigma(W_1x + b_1) + b_2 \; ; \; MAC_{FFN} \approx 2Tdf$$

Softmax demands exponentiation or approximation, a summation reduction and reciprocal/division. Typical LayerNorm computes means, variances and reciprocal square roots. Lookup-table approximations, fixed-point

arithmetic and alternative normalization can lower cost, but must be checked for changes in logical error rate. The feedforward network can dominate total arithmetic for very short sequences.

2.4 Worked arithmetic count: five-token Transformer

For T = 5, d = 16, H = 1, f = 32, one encoder block (not counting input embedding, softmax, normalization or output classifier):

Operation	Expression	MAC terms
Q/K/V and output projection	$4 \times 5 \times 16^2$	5,120
QK^T score product	$5^2 \times 16$	400
Attention-weighted values	$5^2 \times 16$	400
FFN dense layers	$2 \times 5 \times 16 \times 32$	5,120
Total	$4Td^2 + 2T^2d + 2Tdf$	11,040

MAC count is workload, not clock cycles. A 120-token sequence with d = 16 has 230,400 attention-score MAC terms versus 400 for five tokens: 576× more score arithmetic, but not necessarily 576× more inference time or total power. Longer tokens also change projections, memory and FFN costs.

3. Hardware parallelism, pipeline depth and timing closure

Parallelism means instantiating more than one arithmetic datapath so independent computations execute simultaneously. A 16-term dot product can reuse one multiply-accumulate unit for 16 products or use 16 multipliers plus an adder reduction tree. The first saves DSPs but usually costs cycles; the second spends resources and routing to lower arithmetic depth.

$$\text{Idealized cycle lower bound} \geq \text{ceil}(N_{\text{independent operations}} / P)$$

P is the number of independent operations served in parallel, not necessarily the raw number of DSP blocks. This lower bound ignores accumulation dependencies, pipeline fill, buffering, memory-port bandwidth and control. If BRAM cannot deliver operands fast enough, extra multipliers may sit idle.

Method	Latency/throughput	Power and resources
Replicate arithmetic	Reduces scheduled cycles when work is independent.	More DSPs, LUTs and active routing.
Reuse arithmetic	Requires more sequential steps.	Fewer arithmetic units; active over a longer interval.
Pipeline stages	Improves II; can increase total L.	More registers and clock-tree switching.
Lower arithmetic precision	May enable smaller and faster circuits.	Potentially less energy per operation; accuracy verification required.

3.1 Clock frequency is not guaranteed

$$T_{\text{critical}} = T_{\text{logic}} + T_{\text{routing}} + T_{\text{setup}} \; ; \; T_{\text{critical}} \leq 1/f_{\text{clk}}$$

Post-place-and-route timing closure depends on the longest register-to-register path. Fan-out, softmax reductions, BRAM read timing, DSP placement and congested routing can reduce f_{max}. Record clock frequency, worst timing slack, exact cycle latency, and II. A simulated 80-cycle decoder at 100 MHz gives 0.8 µs only if the routed hardware meets the 100 MHz clock.

4. Complete-window decoding versus streaming decoding

4.1 Full-window Transformer

Collect all detector data required by a finite Stim experiment, then run the model once. This matches final observable-flip labels for a complete memory experiment and simplifies training/evaluation. The output is unavailable until all necessary measurements, including terminal events where applicable, have arrived.

$$T_{\text{result from start}} = T_{\text{collect window}} + T_{\text{transport}} + L + T_{\text{deliver}}$$

4.2 Streaming Transformer

Update the decoder each time a new round arrives, using a fixed rolling history, cached state, or causal attention. Streaming may avoid repeated work and improve responsiveness but increases state-management and buffering complexity. Targets must match information available at each inference time: do not use a complete experiment's final observable flip as the correct action for all prior rounds.

For a rotated distance-5 code, there are 24 ordinary stabilizer measurements each round. Yet the full Stim distance-5, five-round memory circuit includes boundary/terminal detector events with distinct spacetime structure; it is not safe to assume that all 120 outcomes form five interchangeable 24-bit temporal chunks. Use Stim detector coordinates and masks for correctly timed token construction.

5. Power consumption at 4 Kelvin

Dissipated FPGA electrical power becomes thermal load that must be removed through the cryostat. The cooling budget available to the decoder is only part of the 4 K stage's nominal capacity. Static FPGA draw, inference logic, I/O, power delivery, cabling and existing equipment must all be accounted for.

5.1 Power model

$$P_{\text{device}} = P_{\text{static}} + P_{\text{dynamic}} \; ; \; P_{\text{dynamic}} \approx \alpha C V^2 f$$

P_{static} covers baseline consumption when programmed and idle; $P_{dynamic}$ depends on circuit activity α, switched capacitance C, supply voltage V and clock rate f. Logic, DSP, BRAM, clock network and I/O contributions may be reported as subdivisions of static/dynamic power; avoid double-counting them.

A published cryogenic Artix-7 experiment measured static power of 95 mW at 300 K and 170 mW at 4 K for its test configuration, showing that ordinary room-temperature estimates cannot simply be extrapolated. Those measurements are not a power prediction for this project's board. [2]

5.2 Cooling budget and die temperature

$$P_{\text{allow,decoder}} \leq Q_{\text{cool,4K}} - Q_{\text{existing}} - Q_{\text{support}} - Q_{\text{margin}}$$

A hypothetical stage with 1 W usable cooling and 0.7 W of existing loads and reserve leaves at most 0.3 W for the decoder system; this is an illustration, not an engineering budget. Even a nominally 4 K mounting plate may have a hotter silicon die because of thermal resistance and local switching hot spots. Measure stage heat load and die/package temperature where possible.

5.3 Transformer operations that dissipate dynamic power

Subsystem	Reason for activity	Architecture response
Q/K/V and FFN	Dense multiplies, adders, accumulation registers.	Reduce embedding/FFN width; consider efficient INT8 hardware.
Attention scores and values	T^2 score products, attention storage and aggregation.	Shorten token sequence; improve data reuse.
Softmax and normalization	Exponentials, reciprocal/division, reductions and approximations.	Evaluate fixed-point LUT and simplified functions.
Memory and routing	Repeated weight/activation transfers, clocked buffers and interconnect.	Keep weights on chip; localize computation; minimize copies.
Control and I/O	Input capture, output signals, state registers.	Use deterministic interfaces; eliminate unnecessary toggles.

5.4 Power versus energy: do not confuse them

$$E_{isolated} \approx P_{active} \times L \; ; \; E_{steady} = P_{average\ at\ target\ rate} / R_{inference}$$

$$\Delta P_{decode} = P_{active} - P_{idle}$$

Multiplying active power by latency is appropriate for a clearly defined isolated active period, but a pipelined decoder has overlapping inferences and continuously consumes static power. Under sustained operation, stage cooling depends on total time-average dissipation at the required prediction rate. Report idle, average full-throughput, and peak/transition current.

Illustrative tradeoff: a 300 mW design completing one isolated prediction in 0.5 µs dissipates approximately 150 nJ over that interval. A 220 mW design lasting 0.9 µs dissipates 198 nJ. The first has higher instantaneous power but lower isolated-interval energy. This does not determine their steady-state energy per prediction.

5.5 Cryogenic power integrity

Long power leads, regulator limitations and low-temperature changes in decoupling capacitors can produce supply-voltage fluctuations. These fluctuations affect gate timing, potentially violating the inference deadline even when nominal average power is low. Homulle et al. describe firmware-based activity stabilization for cryogenic FPGAs [3]. Therefore evaluate supply ripple, current peaks and timing stability along with average power.

6. Architecture selection under the two constraints

$$\min P_{4K}(\text{model, hardware}) \text{ subject to } L \le 1\ \mu s,\ II \le T_{round},\ LER \le LER_{limit}$$

The model architecture sets arithmetic workload and information capacity; hardware scheduling/parallelism determines clock cycles, throughput, resource use and switching. The target is the lowest-power feasible implementation that meets the timing requirement with margin while maintaining acceptable logical error rate against PyMatching.

Architecture variable	Why it changes latency	Why it changes power
1 versus 2+ encoders	Additional serial dependent layers.	More arithmetic, state and data transfer.
Embedding d = 8,16,32	Many projection/FFN MACs scale near d^2.	More active arithmetic and larger weights.
3–5 round tokens vs detector tokens	Attention scales as T^2.	Scores, buffers and read/write traffic increase.
1 vs 2 heads at fixed d	Similar leading arithmetic, different scheduling.	More control/buffer overhead possible.
FFN f = 16,32,64	Cost ≈ 2Tdf MAC terms.	Often meaningful even with few tokens.
INT8 / INT4 fixed point	May enable shorter hardware pipelines.	Less storage and arithmetic cost if implemented efficiently.

Design example. Begin with one encoder, embedding 16, one head, FFN 32 and five correctly grouped syndrome-round tokens, then compare with 8- and 32-wide variants. Train FP32, check INT8/INT4 behavior, synthesize the hardware, adjust multiplier replication/reuse and resynthesize until both timing and power goals are evaluated. None of these settings guarantees 1 µs or superior decoding accuracy.

7. Stim generation, training batches and hardware validation

7.1 Training batch size is not FPGA inference batch size

The notebook’s last two pages cover batch size, optimizer steps and activation memory. During training on a GPU, B independent Stim-generated shots are processed together before a parameter update. On the FPGA, low-latency decoding normally evaluates one current syndrome/window per inference. A training batch of 128 does not imply the FPGA must infer on 128 shots at once.

$$X_{batch} \in \{0,1\}^{B\times T\times m} \; ; \; g_B = (1/B)\,\Sigma_{i=1..B}\,\nabla_\theta\,\square_i$$

$$\text{Steps per epoch} = \text{ceil}(N_{shots}/B) \; ; \; N=1000,B=16 \rightarrow 63 \; ; \; N=1000,B=128 \rightarrow 8$$

Larger training batches can require more activation memory. A Transformer stores projection activations, attention intermediates and FFN activations during backpropagation; an RNN stores time-step activations; a CNN retains intermediate feature maps. Parameters, gradients and optimizer state add memory overhead. Training batch size influences GPU usage and optimization behavior, not the FPGA’s single-shot inference latency.

7.2 Fair model comparison and hardware measurement

$$LER = (1/N)\,\Sigma\,1[\hat{y}_i \neq y_i] \; ; \; SE(LER) \approx \text{sqrt}(LER(1-LER)/N)$$

Generate independent train/validation/test shots at specified Stim distances, rounds and physical error rates. Compare model predictions against the same observable-flip labels and PyMatching on identical held-out shots. Class-imbalance can make conventional accuracy misleading; logical error rate and its uncertainty are more relevant. The quoted standard-error expression is a Bernoulli approximation for independent shots.

Stage	What it establishes
Stim + PyMatching	Decoder accuracy under the simulated noise model, not FPGA power or timing.
Quantized bit-exact inference	Quantization impact, overflow and agreement with the intended hardware arithmetic.
HLS/RTL simulation	Cycle latency, II, functional output and signal activity (VCD/SAIF).
Synthesis and place-and-route	Routed fmax, worst slack, DSP/LUT/FF/BRAM use and critical path.
Power estimation and cryogenic testing	Switching-based estimate, active/idle rail power, heat load and voltage stability at 4 K.

Use representative Stim syndrome streams at the intended update frequency to generate switching traces. Rare detector firings do not automatically save power: conventional dense circuitry may continue to clock and multiply regardless of input sparsity. A power advantage from sparsity requires operand isolation, gating or a sparse datapath in the FPGA design.

8. Research references

[1] Jiang, Z. et al. “Low latency transformer inference on FPGAs for physics applications with hls4ml.” JINST (2025), arXiv:2409.05207. Transformer attention, normalization and FPGA implementations. https://arxiv.org/abs/2409.05207

[2] Conway Lamb, I. D. et al. “An FPGA-based instrumentation platform for use at deep cryogenic temperatures.” Rev. Sci. Instrum. 87, 014701 (2016). Artix-7 power and clock characterization at 4 K. https://doi.org/10.1063/1.4939094

[3] Homulle, H., Visser, S., Patra, B. and Charbon, E. “Design techniques for a stable operation of cryogenic field-programmable gate arrays.” Rev. Sci. Instrum. 89, 014703 (2018). https://doi.org/10.1063/1.5004484

[4] Caune, L. et al. “Demonstrating real-time and low-latency quantum error correction with superconducting qubits.” Nature Communications 17, 7383 (2026). Sub-µs mean decoding per round and separate full-feedback latency. https://doi.org/10.1038/s41467-026-73331-6

[5] Agostinelli, V., Agostini, N. B. and Tumeo, A. “UltraFormer: An Efficient Transformer for FPGAs.” IEEE FCCM (2025). Low-precision and efficient-attention FPGA methods. https://doi.org/10.1109/FCCM62733.2025.00034

[6] Quantumlib. “Stim detector error model format.” Technical documentation on detector and logical observable targets and spacetime coordinates. https://github.com/quantumlib/Stim/blob/main/doc/file_format_dem_detector_error_model.md

Boundary of evidence. Demonstrations of FPGA Transformers, sub-microsecond QEC decoding and cryogenic FPGA operation are separate results. Their combination does not prove that this particular Transformer will satisfy 1 µs at 4 K; that requires post-route timing, actual rail/heat measurements and verified decoding quality.